# Computer Communication and Networks

## Network Layer: Introduction to Routing

Session 15

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# Network Layer

**Learning Outcome:-** At the end of the lesson, you are expected to be able to:

1. Introduction to Routing
2. Types of Routing
3. Inter VLAN routing
4. Layer-3 switches configuration



# Network Layer: Routing

## Introduction

- Routing is the process of determining the best path for data to travel in a network.
- Routers forward packets based on destination IP addresses.

- **Function of Routing**
  - Path Determination: *Selects optimal path.*
  - Packet Forwarding: *Sends data to the destination.*
  - Network Segmentation: *Breaks networks into different broadcast domains.*





# Routing Protocol

- The systems used on the Internet are so large that one routing protocol cannot handle the task of updating the routing tables of all routers.
  - Therefore, the Internet is divided into **Autonomous Systems (AS).**
    - An autonomous system is a **group** of networks and routers under the authority of a single administration.
- Routing inside an Autonomous System is referred to as **intra-domain routing.**
- Routing between autonomous systems is referred to as **inter-domain routing.**



# Network Layer: Routing Protocols

## Routing Protocol: Intra-domain routing Protocol

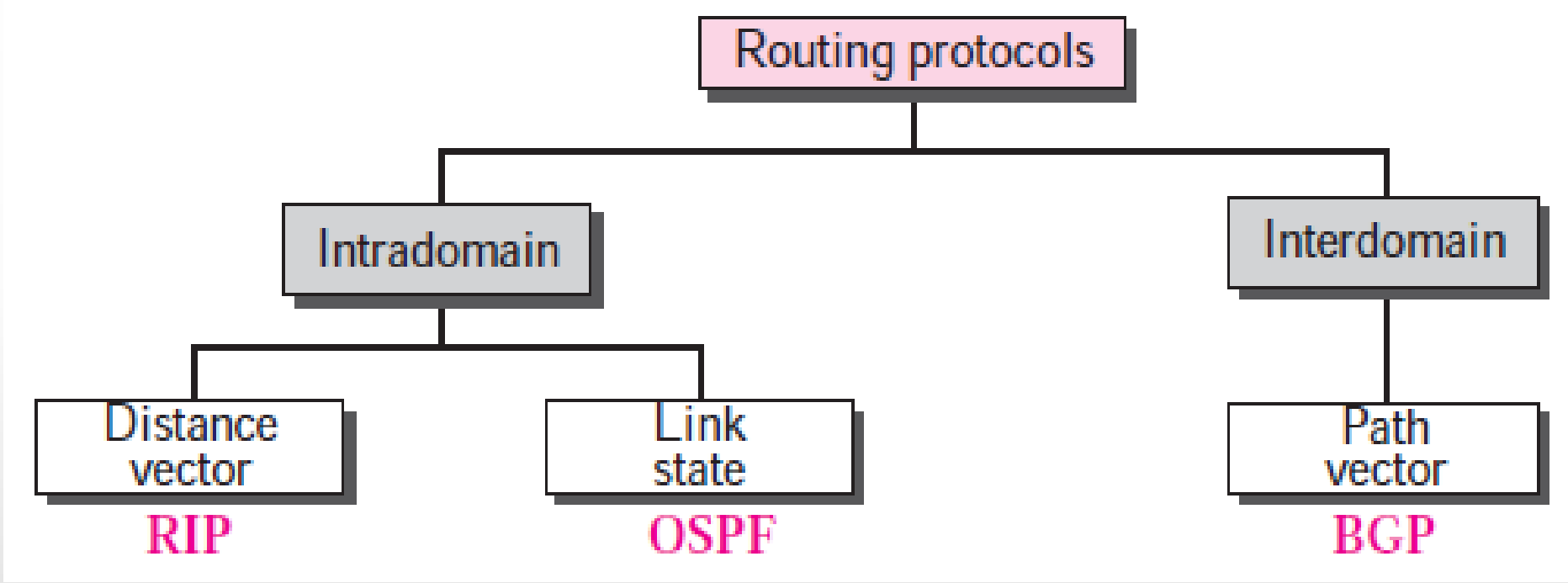
- **Intra-domain Routing Protocol**
  1. Distance Vector Protocol → RIP
  2. Link State Protocol → OSPF
     - RIP and OSPF are the interior routing protocols

- **Inter-domain Routing Protocol**
  1. Path Vector →BGP
     - BGP is an exterior routing protocol



# Network Layer: Routing Protocols

## Routing Protocol: Routing Protocols







# Introduction

- **Types of Routing Protocol**
  1. Static Routing
  2. Dynamic Routing
  3. Default Routing

- Manually configured routes.
- Pros: Simple, no overhead, full control.
- Cons: Not scalable, manual updates required.



# Network Layer: Routing

## Introduction

- **Types of Routing Protocol**
  1. Static Routing
  2. Dynamic Routing
  3. Default Routing

- ❖ Routers learn routes automatically.
- ❖ Pros: Adapts to changes, scalable.
- ❖ Cons: More CPU and memory usage, complex setup.

- ❖ Protocols
1. Distance Vector Protocols (e.g., RIP, IGRP)
2. Link State Protocols (e.g., OSPF, IS-IS)
3. Hybrid Protocols (e.g., EIGRP)





# Introduction

- **Types of Routing Protocol**
  1. Static Routing
  2. Dynamic Routing
  3. Default Routing

- ❖ Routes packets when no specific route is available.
- ❖ Commonly used for unknown destinations.
- ❖ Pros: Simplifies routing tables.
- ❖ Cons: May lead to inefficient paths.



# Network Layer: Routing

## Introduction

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## Distance Vector Protocols

- Use hop count as the metric.
  - Examples: RIP, IGRP.
- Advantage
  - Easy configuration, good for small networks.
- Disadvantage:
  - Slow convergence, potential routing loops.





## Link State Protocols

- Builds complete map of network topology
  - Examples: OSPF, IS-IS
- Advantage
  - Faster convergence, more accurate.
- Disadvantage:
  - More complex, higher resource consumption.



# Network Layer

## VLAN

- A Virtual Local Area Network (VLAN) is a logical grouping of network devices that are segmented based on function, department, or other factors, without regard to their physical location.
  - Network segmentation for better performance.
  - Enhanced security as broadcast traffic is contained within the VLAN.
  - Easier network management and isolation of broadcast domains.





## Inter-VLAN Routing

- The process of routing traffic between different VLANs.
- Since VLANs are separate broadcast domains, devices in one VLAN cannot communicate with devices in another VLAN without a Layer-3 device (such as a router or Layer-3 switch)
  - Layer-3 Switch Configuration ←



# References

## References:


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2. Tanenbaum, Computers Networks, PHI
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4. Forouzan, B.A., TCP/IP Protocol Suit, 4th Edition.
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See you in next class.